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Real-Time Diabetes Detection Using Machine Learning and Apache Spark

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***Abstract*— The convergence of contemporary and cutting-edge technologies, including big data analytics and machine learning, is reshaping the scale and efficiency of healthcare systems. This work presents distinct approaches and contributions, addressing specific challenges in healthcare and disease and diabetes analytics. The real-time healthcare for disease diabetes and the integration of big data analytics, machine learning, and real-time processing have paved the way for innovative solutions to address disease diabetes prediction and monitoring. It's explored innovative solutions to overcome challenges in healthcare analytics, offering real-time predictions and continuous monitoring for improved patient care. The real-time Healthcare-Diabetes dataset was analyzed using various machine learning models, and processing in real-time has led to innovative solutions to address diabetes prediction and monitoring. The Gradient Boosted Tree Classifier emerged as the most accurate model with an accuracy of 90.14%, followed by the Decision Tree Classifier at 84.62%, the Random Forest Classifier at 82.84%, the Linear Support Vector Classifier at 78.70%, and Logistic Regression at 64.69%. These results demonstrate the system's robustness and efficiency in real-time data collection, processing, and prediction. Leveraging Apache Spark and open-source big data technologies, specify data challenges and advocate for scalable, efficient, and cost-effective healthcare analytics. It contributes to the ongoing transformation of healthcare systems, demonstrating the effectiveness of advanced technologies in enhancing disease prediction, monitoring, and overall healthcare services.**

Keywords*— *Healthcare analytics, real-time prediction, Apache Spark, machine learning, distributed computing, personalized healthcare.

I. INTRODUCTION

The cutting-edge of technology, such as big data analytics and machine learning, is revolutionizing the healthcare industry. For efficient therapy and patient care, a timely and precise identification of the condition is essential. This work presents a real-time illness Diabetes diagnosis prediction system intended to tackle the difficulties associated with medical diagnostics. To improve the speed and precision of illness prediction, we used to integrate Apache Spark's [1] parallel processing capabilities with machine learning methods. Diseases, particularly chronic ones like diabetes, pose huge challenges for both individuals and healthcare systems worldwide. The need for timely and precise illness Diabetes diagnosis, monitoring, and prediction have never been better. As a result, data mining is a powerful technique for extracting insights from enormous databases. Diabetes is a chronic disease that affects millions of people worldwide and is getting increasingly prevalent [2].

Accurate and timely diabetes diagnosis is hindered by traditional methods, which are slow and often inaccurate, affecting patient outcomes. There is a critical need for a scalable, real-time system that leverages advanced machine learning and big data technologies to provide precise and immediate predictions. This study addresses these challenges by developing a robust real-time diabetes diagnosis prediction system using Apache Spark, Kafka, and Spark Streaming, aimed at improving the accuracy and speed of diabetes detection.Diabetes is regarded as the disease of the present century due to a lack of physical exercise, a bad diet, and hereditary predisposition. The global diabetes population is believed to be in the millions, and this figure is growing year after year [3]. Diabetes, also known as hyperglycemia, develops when blood glucose levels rise or the body's cells fail to produce enough insulin to absorb sugar in the cells.This can cause arterial wall damage, raise the patient's risk of atherosclerosis, and impair vision [4].Current systems struggle with data volume and pace. To solve this issue, a real-time diabetes diagnostic prediction system based on Apache Spark and machine learning is presented. This system processes and analyses healthcare data, providing rapid and accurate projections via complex predictive algorithms, allowing medical workers to take preventative measures.

The project's goal is to create a real-time diagnostic system for disease prediction that uses machine learning techniques and Apache Spark to handle data efficiently.The scope covers integrating Apache Spark with machine learning techniques, handling healthcare data, and evaluating its efficacy in a healthcare setting.This enables speedier decision-making, personalized therapy, and better patient results. Predictive algorithms can detect disease symptoms early, but a centralized server architecture optimizes data storage, retrieval, and processing, enhancing scalability, security, and accessibility when dealing with health-related data from several sources. One of the difficulties in dealing with a diabetes diagnosis is the administration of clinical trials or blood testing [5]. Diabetes latency occurs when the condition is not diagnosed in a timely manner. Its symptoms appear when it has caused diverse organ damage in the body. Diabetes has a number of early warning indicators that researchers might use to diagnose it. For example, people's weight and age are important indications in detecting this hazardous disease, and other methods such as data mining and machine learning can aid in early detection [6].

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II. RELATED WORK

The study by Singh, P., Singh, N., Singh, K.K., & Singh, A.2021. [7] provided an overview of recent advances in machine-learning algorithms and their significant influence on disease detection and diagnosis. It began by highlighting issues with traditional healthcare systems before delving into various machine-learning algorithms like SVM, KNN, Naive Bayes, and Decision Tree. The practical applications of these algorithms in Python were investigated, with a particular emphasis on their use in diagnosing diseases such as cancer, diabetes, epilepsy, and heart attacks. The study included a discussion of the conceptual and mathematical aspects of accuracy, precision, recall, and F1 score metrics used to assess the performance of machine-learning algorithms in disease diagnosis. Finally, the effect of machine learning on the healthcare industry is investigated.

Ahmed, H., Younis, E. M., & Ali, A. A.2020.[8] This study aims to develop a predictive system for diabetes using distributed machine learning on big data platforms like Apache Spark. Five classification techniques were used, with LR achieving the highest accuracy at 82%, followed by LR at 78% and DT at 67%, and SVM at 64%.

Abdali-Mohammadi, Fardin, Maytham N. Meqdad, and Seifedine Kadry ,2020 . [9] Disease diagnosis system built on the foundation of the Internet of Things. Wearable sensors recorded the patient's physiological signals, which were then transmitted to a server on the network. Furthermore, this paper proposed a novel hybrid decision-making method for diagnosis. This approach starts with the creation of a feature set derived from patient signals, which is then processed by a learning model. The final diagnosis was performed using a neural fuzzy model. To evaluate the system's effectiveness, specific disease diagnoses, such as identifying normal and abnormal pulse patterns or diagnosing diabetic issues, are simulated. A disease diagnosis system based on IoT sensors and a hybrid decision-making technique has shown promise in simulating specific disease diagnoses such as identifying abnormal pulse patterns and diagnosing diabetes.

Kumar, Priyan Malarvizhi,et al,2018. [10] proposed a novel Cloud and IoT-based Mobile Healthcare application. This framework, created for public use, takes a systematic approach, focusing specifically on diabetes. The relevant medical data is derived from the UCI Repository dataset, and medical sensors are used to predict severe cases of diabetes. They also introduced a new classification algorithm called the Fuzzy Rule-based Neural Classifier for disease diagnosis and severity assessment. Experiments were carried out using the standard UCI Repository dataset and real health records obtained from various hospitals. The results of these experiments show that our proposed approach outperforms existing systems in disease prediction.

Hussein,A.S.,Omar,W. M.,Li, X.,&Ati, M. 2012. [11] Recommender systems in healthcare for Chronic Disease Diagnosis (CDD) introduced to play a significant role in disease control by offering precise and reliable predictions of disease risk diagnosis and recommendations for medical advice. Decision tree algorithms are the foundation of multiple classifications used to build an accurate predictive model for disease risk diagnosis in monitored cases. The Unified Collaborative Filtering method, which involves learning a classification model from historical binary ratings and external features, is used to improve recommendation accuracy when providing medical advice to patients.An Attribute Selection method is used to identify relevant features, which reduces data generation while improving predictive model performance. The integration of patients' lab and home test results is thought to improve diagnostic accuracy. A diabetes diagnosis case study designed as an experiment to demonstrate the model's feasibility. The challenge is to provide accurate real-time recommendations for medical data given its complexity, which includes imbalances, large datasets, multidimensional information, noise, and/or missing data. The CDD system was designed to provide high accuracy in disease risk prediction and medical advice recommendations.

Shah, Krish, et al,2020. [12] The authors work Diabetes is a chronic disease with increasing deaths. The Naïve Bayes Classifier predicts diabetes with 81% accuracy. The system allows for live streaming of data input and has the highest classification accuracy of 0.808. The Gaussian Naïve Bayes Classifier outperformed the Multinomial Naïve Bayes Classifier. The research shows that Gaussian Naïve Bayes Classifier outperforms Multinomial Naïve Bayes in predicting diabetes, as evidenced by performance evaluation metrics like accuracy, precision, recall, and F-Measure.

Ibrahim,Mohamed Said, and Sameh Saber,2023.[13] Academic investigation delves into the influence of machine learning and predictive analytics on healthcare's disease prevention. These technologies have the potential to revolutionize disease identification, diagnosis, and treatment processes by analyzing large amounts of patient data, thereby improving preventive measures. The study focuses on specific domains where machine learning and predictive analytics are important. Machine learning algorithms' ability to discern subtle patterns within medical records, laboratory results, genetic information, and imaging data is particularly useful for early detection and diagnosis. Such algorithms make a significant contribution to the early detection and diagnosis of conditions such as cancer, cardiovascular disease, diabetes, and neurodegenerative disorders.

Saravi, Farshid Bagheri, et al 2021. [14] employs machine learning (ML) techniques in the Apache Spark computing environment to diagnose diabetes in real time. Large-scale medical samples are analyzed using Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM) techniques. The results show that using these strategies reduces runtime, with non-distributed systems and Apache Spark systems taking 43.40 and 6.98 seconds, respectively. The Apache Spark framework can improve disease diagnosis accuracy while reducing runtime. A future study proposes connecting WBAN with Apache Spark technology to monitor hospital patient status.

Alfian, Ganjar, et al,2018.[15] study Alfian, Ganjar, et al, proposes a personalized healthcare monitoring system that uses BLE-based sensors and real-time data processing, combined with machine learning algorithms for diabetes prediction and blood glucose level forecasting, demonstrating potential for early detection and self-management of diabetes.

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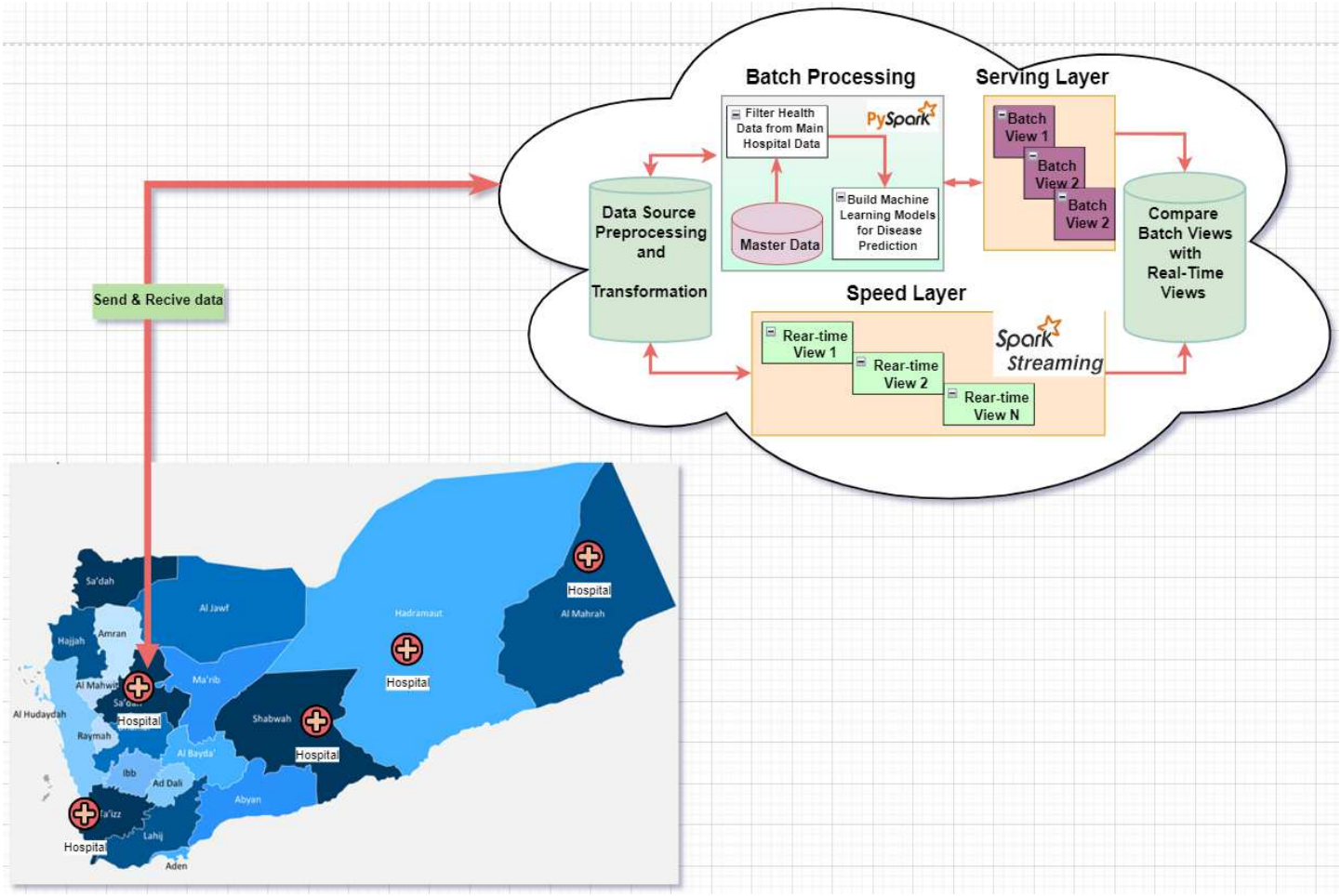

Figure 1: Proposed System Architecture

III. ROPOSED SYSTEM ARCHITECTURE

The real-time disease diagnostic prediction system's construction and implementation are described in full, step-by-step, in the methodology section. The integration of big data analytics, Apache Spark, and machine learning requires the effective completion of each step in the process.

A. Methodology

The proposed real-time diabetes diagnostic prediction system combines big data analytics with Apache Spark and machine learning.We use a dataset that has been collected from multiple hospitals, cleaned, and transformed into a standard format. The data is processed in batches with PySpark to create and optimise machine learning models for diabetes prediction. Spark Streaming processes real-time data to generate immediate health insights, whereas batch views analyse historical health data. The system uses Lambda Architecture to efficiently handle both real-time and historical data, resulting in accurate and timely diabetes diagnoses.

B. System Description

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

1. Data Acquisition: Send & Receive Data from Hospitals

Data is gathered from multiple hospitals at the beginning of the process. This entails setting up a reliable system for data transmission and reception pertaining to health. For the forecasts to be accurate and timely, a reliable and effective data transfer system must be in place.

2. Data Source Preprocessing and Transformation

It undergoes preprocessing and transformation. Cleaning the data, handling missing values, and transforming it into a suitable format for analysis. The goal is to ensure that the data is standardized and ready for both batch and real-time processing.

3. Batch Processing (PySpark)

Batch processing is the process of performing several data processing operations on a collection of data (a batch) all at once. Handles data in chunks or batches after a set amount of time or upon reaching a specific size, in contrast to real-time or stream processing, which processes data as it comes in.Using Spark's capabilities to process and analyse massive amounts of data in a distributed and parallelized fashion is known as batch processing in the context of PySpark, the Python API for Apache

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Spark.PySpark offers a Python programming interface for creating Spark applications, use Spark's central data processing engine to carry out calculations on distributed datasets.

3.1 Filter Health Data from Main Hospital Data: PySpark is used to filter health data from the primary hospital dataset during the batch processing stage. In order to create a master dataset and choose pertinent attributes and records needed for training machine learning models.

3.2 Build Machine Learning Models for Disease Prediction Within the batch processing layer, machine learning models are constructed using PySpark. These models are trained on historical data to predict diseases accurately. The focus is on optimizing and fine-tuning these models to enhance prediction accuracy.

3.3 Master Data: The master dataset, derived from the batch processing layer, serves as a comprehensive repository of preprocessed and transformed health data. It acts as a foundational element for both real-time and batch views.

4. Speed Layer (Spark Streaming)

In the speed layer, Spark Streaming is employed for real-time data processing. This phase involves the creation of multiple real-time views, each providing instantaneous insights into different aspects of health data.

Real-time View 1, 2, n:Multiple real-time views are generated, offering immediate insights into different dimensions of health data. This includes monitoring vital signs, anomaly detection, and other critical parameters.

5. Serving Layer

The serving layer is responsible for creating batch views, which are comprehensive snapshots of historical health data.

Batch View 1, 2, n:Batch views are created to provide a holistic overview of historical health data. These views assist in analyzing trends, patterns, and long-term insights.

6. Query

The query phase involves comparing batch views with real-time views to ensure consistency and accuracy. Compare Batch Views with Real-Time Views: By querying and comparing batch and real-time views, the system validates the accuracy of real-time predictions against historical data. This step is crucial for evaluating the effectiveness and reliability of the entire prediction system.

C. *Lambda Architecture*

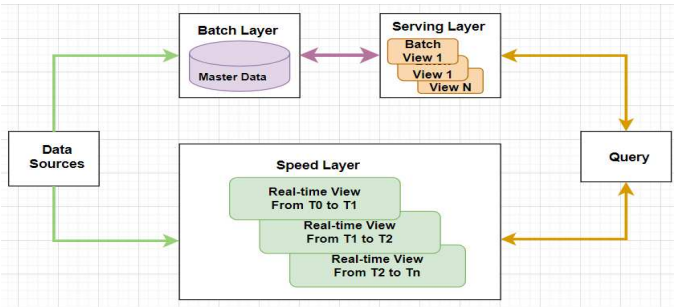

Figure 2:Lambda Architecture

The suggested architecture employs Lambda Architecture concepts to efficiently process both historical and real-time data. The Batch Layer processes historical data in big batches to provide insights into healthcare analytics [18].The Speed Layer conducts real-time data processing, allowing for rapid response to changing situations. The Serving Layer combines results from both layers to provide a comprehensive perspective of healthcare patterns and patient situations. This design enables healthcare providers to make informed decisions about patient care while also providing a smooth user experience. The Lambda Architecture enables the system to manage complicated healthcare data, providing a comprehensive and responsive solution for healthcare analytics and disease prediction.

D. *Apache Spark*

Apache Spark is an open-source, distributed computing system that offers a quick and versatile cluster computing platform for big data processing and analytics.

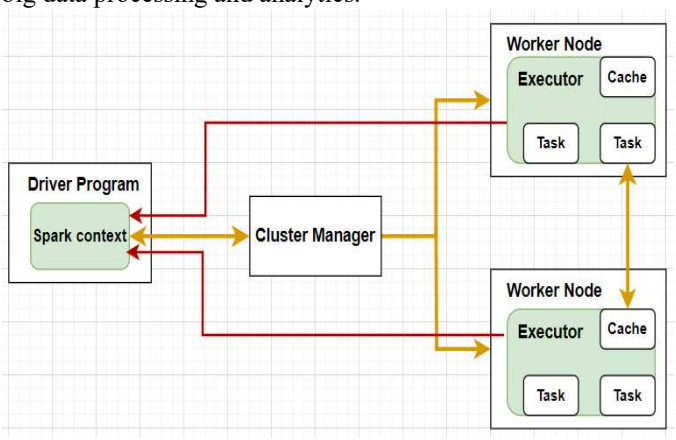

Figure 3:Apache Spark

It was created to address the limits of the Hadoop MapReduce model, providing a more adaptable and efficient alternative. Apache Spark is a prominent open-source platform for large-scale data processing, particularly for iterative machine learning tasks. Spark is a fault-tolerant, general-purpose cluster computing system having APIs in Java, Scala, Python, and R, as well as an optimised engine that enables broad execution graphs. Furthermore, Spark is efficient at iterative computations, making it ideal for the creation of large-scale machine learning applications [16].

E. *Spark Streaming*

Spark Streaming is an extension to the main Spark API that allows for scalable, high-throughput, fault-tolerant stream processing of live data streams. Spark Streaming no longer receives updates and is considered an obsolete project. Structured Streaming is Spark's updated and more user-friendly streaming engine. Spark Structured Streaming is best suited for streaming applications and pipelines. Data can be ingested from a variety of sources, including Kafka, Kinesis, and TCP connections, and then processed using complicated algorithms represented as high-level functions like as map, reduce, join, and window. Finally, processed data can be distributed to

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filesystems, databases, and live dashboards. Spark Streaming employs the discretized stream (D-Streams) paradigm to organise stream processing as a sequence of stateless, deterministic batch computations across short time intervals, resulting in high throughput and fault tolerance[17]. Spark streaming applications are long-running, nonterminating stream processing services that are managed by a Spark driver.

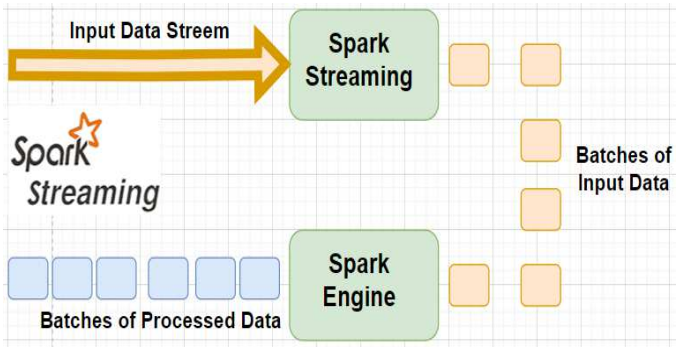

Figure 4:Spark Streaming

F. Datasets description and analysis

The dataset part has a significant impact on the real-time disease diagnosis prediction system, which uses Apache Spark and Machine Learning. This section examines and introduces the two key datasets that serve as the basis for our analytical and predictive modelling work. These datasets must be employed in the training and testing phases of machine learning models so that our system can provide precise and fast predictions for healthcare diagnostics.

Table 1: Healthcare Diabetes dataset

Attribute	Description
Id	Unique identifier for each data entry.
Pregnancies	Number of times pregnant.
Glucose	Plasma glucose concentration .
BloodPressure	Diastolic blood pressure (mm Hg).
SkinThickness	Triceps skinfold thickness (mm).
Insulin	2-Hour serum insulin (mu U/ml).
BMI	Body mass index (weight in kg / height in m^2).
DPF	Diabetes pedigree function.
Age	Age of the patient.
Outcome	Binary classification (1)or absence (0) of diabetes.

Healthcare Diabetes Dataset: The Healthcare Diabetes Dataset, our second dataset, is an extensive compilation of health-related characteristics created to aid in the creation of predictive models for determining who is at risk of developing diabetes. Including characteristics like blood pressure, glucose concentration, and genetic scores, this dataset includes a variety of markers that are essential for determining the risk of diabetes [19]. Following ethical protocols and taking privacy concerns into account, we use this dataset to find trends and insights that help with risk assessment accuracy and customized treatment plans.The histogram for the "Age" column in the healthcare dataset displays the distribution of ages among the surveyed individuals. The x-axis represents the age ranges, and the y-axis indicates the frequency or count of individuals falling within each age category. The histogram provides a visual representation of the age distribution, highlighting any prevalent age groups or patterns within the dataset.

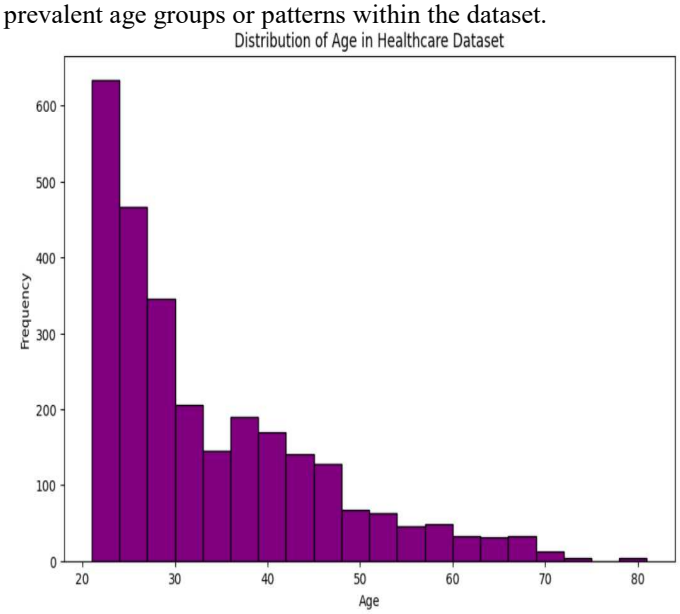

Figure 5:Spark Streaming

IV. RESULTS AND DISCUSSION

The Healthcare-Diabetes dataset was analyzed using a variety of machine learning algorithms, and the results showed promise in predicting diabetes outcomes. This section provides a comprehensive assessment of the performance of several models, such as Logistic Regression, Decision Tree Classifier, Random Forest Classifier, Linear Support Vector Classifier (Linear-SVC), and Gradient Boosted Tree Classifier (GBT). Each model's accuracy in predicting diabetes outcomes is evaluated and compared, providing useful information on the usefulness of various diabetes prediction algorithms. The Healthcare-Diabetes dataset was analyzed using various machine learning models, with Gradient Boosted Tree Classifier showing the highest accuracy (GBT Classifier) (90.14%), Decision Tree Classifier (84.62%), Random Forest Classifier (82.84%), Linear Support Vector Classifier (LinearSVC) (78.70%), and Logistic Regression (64.69%) . In comparison, Ahmed, H., Younis, E. M., & Ali, A. A.2020 [8] reported the highest accuracy for Logistic Regression at 82%, while Decision Tree, Random Forest, and Support Vector Machine models achieved lower accuracies of 67%, 78%, and 64% respectively. Notably, our study demonstrated superior performance for Decision Tree and Random Forest classifiers, whereas Logistic Regression performed better in Ahmed et al.'s study.These models showed high accuracy in predicting outcomes, providing insights into diabetes prediction. These findings have significant implications for diabetes prediction and prevention efforts in healthcare. By leveraging machine learning algorithms and predictive modelling techniques, healthcare providers can identify individuals at risk of developing diabetes more accurately and efficiently. This proactive approach allows for timely interventions and personalized care strategies, ultimately improving patient outcomes and reducing the burden of diabetes-related complications.

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Table 2:Result Accuracy

Models	Accuracy
Logistic Regression	64.69%
Decision Tree	84.62%
Random Forest	82.84%
Linear Support Vector	78.70%
Gradient Boosted Tree	90.14%

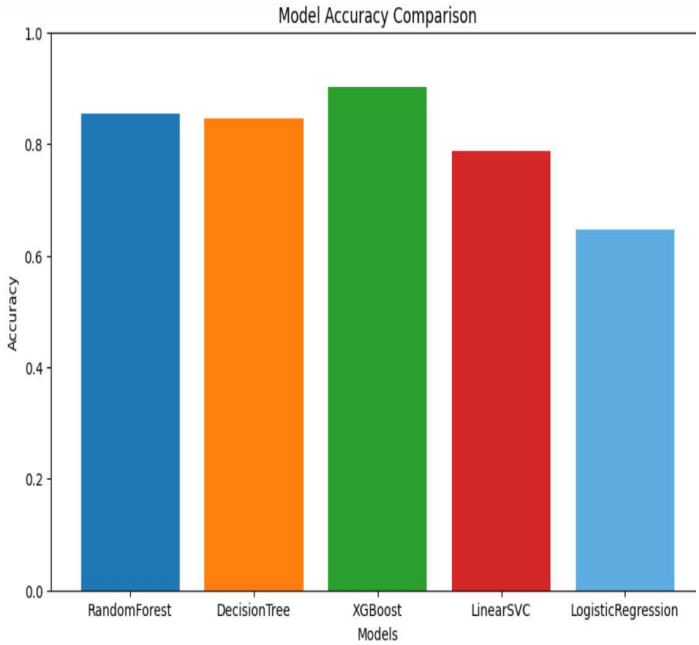

Figure 6:Spark Streaming

V. Conclusion

The real-time illness a diabetes diagnosis prediction system that utilises Apache Spark and machine learning technologies to address the challenges of timely and precise disease identification in the healthcare industry. The system combines Apache Spark MLlib and Spark Streaming, offering a scalable and distributed framework for managing and predicting illnesses like diabetes in real-time. The system's effectiveness is validated through a series of experiments, demonstrating its effectiveness in data collection, real-time data processing, machine learning model evaluation, Spark Streaming performance, integration, scalability, fault tolerance, and forecast accuracy. The proposed system signifies a step forward in healthcare diagnostics, providing a foundation for timely and precise disease identification. Future research directions include incorporating additional healthcare data sources, exploring real-time interactions with user requests, and integrating with healthcare providers for a complete real-time healthcare system. In conclusion, the real-time illness The diabetes diagnosis prediction system presented in this work is at the forefront of healthcare innovation, offering an efficient, scalable, and proactive solution for the healthcare industry. The proposed continuous diabetes disease monitoring system leverages Apache Spark's Resilient Distributed Datasets (RDDs) and machine learning libraries to support in-memory data storage and distributed computing. The work demonstrates a shift towards real-time prediction, providing continuous monitoring for diabetes patients.

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